Elliptic Curves

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To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 1

Introduction To Elliptic Functions

Starting with an ellipse defined by the equation $(x/a)^2 + (y/b)^2 = 1$, a natural question of knowing what is the perimeter of the ellipse is. By consider the equation on the first quadrant, we have that it can be parameterized with the curve

$$y = f(x) = b\sqrt{1 - \frac{x^2}{a^2}}$$

using this, the symmetry of the ellipse and some changes of variables, we have that the formula for the ellipse perimeter is as follows

$$L = 4\int_0^a \sqrt{1 + f'(x)}\,\mathrm{d}x = 4\int_0^1 \sqrt{\frac{1 - k^2t^2}{1 - t^2}}\,\mathrm{d}t$$

notice that the integral is path dependent since $\sqrt{\cdot}$ is double-valued, thus we make branch cuts in the values where both the nominator and denominator vanish so $z \in \{\pm 1, \pm 1/k\}$. By taking two copies of $\mathbb{P}^1_{\mathbb{C}}$ and identifying the branch cuts we get a torus. Thus study of the arc-length of an ellipse leads naturally to the study of integrals on a torus which is an elliptic curve.

1.1. Lattices & Complex Tori

Definition 1.1.1 (Lattice): *Let $w_1, w_2 \in \mathbb{C}$, the complex torus is defined as the set $\Lambda = w_1\mathbb{Z} + w_2\mathbb{Z}$ where $\Lambda \otimes \mathbb{R} = \mathbb{C}$.*

Note: *We define that $\Lambda \otimes R = w_1R + w_2R$.*

An equivalent definition we can get is to consider w_1, w_2 to be linearly independent on $\mathbb{C}$ as an $\mathbb{R}$-vector space.

Definition 1.1.2 (Fundamental Parallelogram): *Let $\Lambda = \langle w_1, w_2\rangle$ be a lattice, the fundamental parallelogram for Λ is a set of the form*

$$D = \{a + \lambda_1 w_1 + \lambda_2 w_2 \mid \lambda_i \in [0, 1)\}$$

Definition (Complex Torus): *Let Λ be a lattice, we define the complex torus associated with Λ as $T := \mathbb{C}/\Lambda$. We can define the addition on the complex torus as $\forall [z_1], [z_2] \in T, [z_1] + [z_2] = [z_1 + z_2]$.*

Note: *To simplify notation, we take that the elements of T are just denoted simply as z instead of $[z]$.*

Proposition 1.1.3: *$(T, +)$ is a commutative group with neutral element $[0]$.*

Definition 1.1.4 (Torsion Element): *Let $z \in \Lambda$, $m \in \mathbb{Z}$, z is said to be of m-torsion if $m \cdot z = 0$. The set of m-torsion elements is denoted $T[m]$.*

Lemma 1.1.5: *$\forall m \in \mathbb{Z}, T[m] \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.*

Proof. Let $\Lambda = \langle w_1, w_2\rangle$, consider the following map

$$\varphi : \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \to T[m], (a, b) \mapsto \frac{a}{m}w_1 + \frac{b}{m}w_2$$

it is well-defined since $m((a/m)w_1 + (b/mw_2) = aw_1 + bw_2 = 0$.

- Let $(a, b), (a', b') \in \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$, $\varphi((a, b) + (a', b')) = \frac{a+a'}{m} + \frac{b+b'}{m} = \frac{a+b}{m} + \frac{a'+b'}{m} = \varphi((a, b)) + \varphi((a', b'))$ so φ is a morphism.
- Let $(a, b), (a', b') \in \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$, $\varphi((a, b)) = \varphi((a', b')) \Rightarrow \frac{a}{m}w_1 + \frac{b}{m}w_2 = \frac{a'}{m}w_1 + \frac{b}{m}w_2 \Rightarrow a = a', b = b' \Rightarrow (a, b) = (a', b')$ so φ is injective.

- Let $\Lambda = \langle w_1, w_2 \rangle$, since $\Lambda \otimes \mathbb{R} = \mathbb{C}$ then $\forall z \in \mathbb{C}, z = aw_1 + bw_2$, then if $z \in T[m]$, $mz = (ma)w_1 + (mb)w_2 \in \Lambda$ thus $ma, mb \in \mathbb{Z}$ hence we obtain $(a, b) = (a'/m, b'/m)$ with $a', b' \in \mathbb{Z}/m\mathbb{Z}$. The map φ is surjective. $\square$

1.2. Elliptic Functions

Definition 1.2.6 (Elliptic Function): *Let $f : \mathbb{C} \to \mathbb{C}$ be a meromorphic function, f is said to be an elliptic function relative to Λ if*

$$\forall z \in \Lambda, \forall x \in \mathbb{C}, f(x + z) = f(x).$$

Definition 1.2.7 (Order Of Vanishing/Residue): *If f admits a Laurent expansion $\sum_{n \geq k} a_k (z - w)^k$ at w then we define*
- *the order of vanishing of f at w as $\operatorname{ord}_w(f) = k$.*
- *the residue of f at w as $\operatorname{res}_w(f) = a_{-1}$.*

Definition 1.2.8 (Pole/Zero): *Let $f : \mathbb{C} \to \mathbb{C}$ be a meromorphic function and $\sum_{n \geq k} a_n (z - w)^n$ its Laurent expansion, then w is called a pole if $k < 0$ and called a zero if $k > 0$.*

Definition 1.2.9 (Order Of Elliptic Functions): *Let $f : \mathbb{C} \to \mathbb{C}$ be an elliptic function relative to Λ, and D be a fundamental domain for Λ. The order of f is defined as the number of poles of f inside of D,*

$$\operatorname{ord}(f) = \sum_{\text{pole } w \in D} \operatorname{ord}_w(f)$$

Theorem 1.2.10: *Let $T = \mathbb{C}/\Lambda$ a complex torus and f an elliptic function relative to Λ, the following statements are true:*
1. *If f has no poles, then it is constant.*
2. $\sum_{w \in T} \operatorname{res}_w(f) = 0$
3. $\sum_{w \in T} \operatorname{ord}_w(f) = 0$.
4. $\sum_{w \in T} \operatorname{ord}_w(f) w \in \Lambda$.
5. *If f has no zeros, then it is constant.*
6. *A non-constant elliptic function has order at least 2.*

Note: *TODO: Proof of the theorem*

Definition 1.2.11 (Function Field): *Let Λ be a lattice, we define the set of elliptic functions relative to Λ as $\mathbb{C}(\Lambda)$, which we call the function field of Λ*

1.3. Divisor Group Of Complex Tori

Definition 1.3.12 (Divisor Group): *Let T be a complex torus, we define the divisor group of T as the free abelian group generated by the points of T*

$$\operatorname{Div}(T) := \bigoplus_{w \in T} \mathbb{Z} \cdot (w)$$

Definition 1.3.13 (Degree Of Divisors): *Let T be a complex torus and $D = \sum n_w (w) \in \operatorname{Div}(T)$, then the degree of D is $\deg D = \sum n_w$.*

Definition 1.3.14 (Degree Zero Divisors): *Let T be a complex torus, we define the subgroup Div^0 of Div as the set*

$$\operatorname{Div}^0(T) = \{D \in \operatorname{Div}(T) \mid \deg D = 0\}$$

Definition 1.3.15 (Divisor Of Elliptic Functions): *Let $T = \mathbb{C}/\Lambda$ a torus and $f \in \mathbb{C}(\Lambda)^\star$ be a non-zero elliptic function with respect to Λ, we define the divisor of f as*

$$\operatorname{div} f = \sum_{w \in T} \operatorname{ord}_w(f) \cdot (w)$$

Lemma 1.3.16: $\operatorname{div} : \mathbb{C}(\Lambda)^\star \to \operatorname{Div}^0(T), f \mapsto \operatorname{div} f$ *is a group morphism.*

Proof. The function is well-defined given that $\deg \operatorname{div} f = \sum_{w\in T} \operatorname{ord}_w(f) = 0$ from the previous theorem, let $w \in \mathbb{C}$, $f(z) = \sum_{n\geq k} a_n(z-w)^n$ and $g(z) = \sum_{n\geq k'} b_n(z-w)^n$, $(fg)(z) = \sum_{n\geq k+k'} \sum_{j=0}^n a_j b_{n-j}(z-w)^n$, $\operatorname{ord}_w(fg) = k + k' = \operatorname{ord}_w(f) + \operatorname{ord}_w(g)$ hence

$$\begin{aligned}
\operatorname{div}\ (fg) &= \sum_{w\in T} \operatorname{ord}_w(fg)\cdot(w) \\
&= \sum_{w\in T} (\operatorname{ord}_w(f) + \operatorname{ord}_w(g))\cdot(w) \\
&= \sum_{w\in T} \operatorname{ord}_w(f)\cdot(w) + \sum_{w\in T} \operatorname{ord}_w(g)\cdot(w) \\
&= \operatorname{div}\ f + \operatorname{div}\ g
\end{aligned}$$

□

Definition 1.3.17 (Summation Map): *Let $T = \mathbb{C}/\Lambda$ be a torus*

$$\operatorname{sum} : \operatorname{Div}^0(T) \to T, \quad \sum n_w(w) \mapsto \sum n_w w \bmod \Lambda.$$

Proposition 1.3.18: *The summation map is a well-defined group morphism.*

Proof. It is clear that the summation map is well defined, consider the divisors $D = \sum n_w(w), D' = \sum m_w(w) \in \operatorname{Div}^0(T)$, then we have

$$\begin{aligned}
\operatorname{sum}\ (D + D') &= \operatorname{sum}\ \left(\sum (n_w + m_w)(w)\right) \\
&= \sum (n_w + m_w) w \bmod \Lambda \\
&= \sum n_w w + \sum m_w w \bmod \Lambda \\
&= \left(\sum n_w w \bmod \Lambda\right) + \left(\sum m_w w \bmod \Lambda\right) \\
&= \operatorname{sum}\ (D) + \operatorname{sum}\ (D')
\end{aligned}$$

Therefore it is a group morphism. □

Theorem 1.3.19: *The following sequence is exact.*

$$1 \longrightarrow \mathbb{C}^\star \longrightarrow \mathbb{C}(\Lambda)^\star \xrightarrow{\operatorname{div}} \operatorname{Div}^0(T) \xrightarrow{\operatorname{sum}} T \longrightarrow 0$$

Proof. Consider the maps $1 \xrightarrow{i_1} \mathbb{C}^\star \xrightarrow{i_2} \mathbb{C}(\Lambda)^\star$ where i_j are the canonical injections since $\{1\} \subset \mathbb{C}^\star \subset \mathbb{C}(\Lambda)^\star$ which are clearly morphisms (by abuse of notation we take $\mathbb{C}^\star$ to be the set of constant elliptic functions)

1. $\operatorname{Im}\ i_1 = \{1\} = \operatorname{Ker}\ i_2$.
2. $\operatorname{Ker}\ \operatorname{div} = \{f \in \mathbb{C}(\Lambda)^\star \mid \operatorname{div}\ f = 0\} = \{f \in \mathbb{C}(\Lambda)^\star \mid \forall w \in T, \operatorname{ord}_w(f) = 0\} = \mathbb{C}^\star = \operatorname{Im}\ i_2$.
3. $\operatorname{Ker}\ \operatorname{sum} = \{D \in \operatorname{Div}^0(T) \mid \operatorname{sum}\ D = 0\} = \{D \in \operatorname{Div}^0(T) \mid \sum n_w w \in \Lambda\}$

Note: *Unfinished*

□

1.4. The Weierstrass $\wp$-Function

Definition 1.4.20 ($\wp$-Function): *Let Λ be a lattice, the Weierstrass $\wp$-function relative to Λ is defined by the series*

$$\wp(z;\Lambda) = \frac{1}{z^2} + \sum_{w\in\Lambda\setminus\{0\}} \left(\frac{1}{(z-w)^2} - \frac{1}{w^2}\right)$$

Proposition 1.4.21: *The first derivative of $\wp$ is*

$$\wp'(z;\Lambda) = -2\sum_{w\in\Lambda} \frac{1}{(z-w)^3}$$

Proof.

$$\wp'(z;\Lambda) = -\frac{2}{z^3} + \sum_{w\in\Lambda\setminus\{0\}} -\frac{2}{(z-w)^3} = -2\sum_{w\in\Lambda} \frac{1}{(z-w)^3}.$$

□

Definition 1.4.22 (Eisenstein Series): *Let $\Lambda \in \mathbb{C}$ be a lattice, the Eisenstein series of weights $2k$ related to Λ is defined as*

$$G_k(\Lambda) = \sum_{w \in \Lambda \setminus \{0\}} w^{-k}$$

Lemma 1.4.23:

- *The series G_{2k+1} is 0 for every $k \in \mathbb{N}$.*
- *The series $G_{2k}(\Lambda)$ is absolutely convergent for $k > 1$.*

Proposition 1.4.24: *Let Λ be a lattice and $\wp = \wp(z; \Lambda)$ the associated Weierstrass $\wp$-function, then we have:*

- *$\wp$ converges uniformly on every compact subset of $\mathbb{C} \setminus \Lambda$.*
- *$\wp$ is an even elliptic function having a double pole with residue 0 at $z = 0$ (by periodicity, this is true for every lattice point in Λ) and no other poles.*
- *$\wp'$ is an odd elliptic curve.*

Proof. The proofs are taken from "*Silverman: The Arithmetic Of Elliptic Curves*".

- Let $z \in \mathbb{C}$, we have for any $w \in \Lambda$ with $|w| > 2|z|$ then

$$\left| \frac{1}{(z-w)^2} - \frac{1}{w^2} \right| = \left| \frac{z(2w-z)}{w^2(z-w)^2} \right| \leq \frac{|z|(2|w| - |z|)}{|w|^2(|w| - |z|)^2} \leq \frac{10|z|}{|w|^2}$$

Note: *Uncomplete, yet again*

□

Theorem 1.4.25: *Let Λ be a lattice and $\wp = \wp(z; \Lambda)$, then we have*

$$\mathbb{C}(\Lambda) = \mathbb{C}(\wp, \wp').$$

Theorem 1.4.26:

- *The Laurent series of $\wp(z)$ at 0 is given by*

$$\wp(z) = \frac{1}{z^2} + \sum_{n \geq 1} (2n+1) G_{2n+2} z^{2n}$$

- *For all $z \in \mathbb{C} \setminus \Lambda$, we have*

$$\wp'(z)^2 = 4\wp(z)^3 - 60G_4\wp(z) - 140G_6$$

Proof.

- The Laurent expansion of $\frac{1}{1-z}$ at 1 is $\sum_{n \geq 0} z^n$. By differentiation we have that

$$\frac{1}{(1-z)^2} = \sum_{n \geq 1} n z^{n-1} \Rightarrow \frac{1}{(1-z)^2} - 1 = \sum_{n \geq 1} (n+1) z^n$$

$$\begin{aligned}
\wp(z) &= \frac{1}{z^2} + \sum_{w \in \Lambda \setminus \{0\}} \left(\frac{1}{(z-w)^2} - \frac{1}{w^2} \right) \\
&= \frac{1}{z^2} + \sum_{w \in \Lambda \setminus \{0\}} \frac{1}{w^2} \left(\frac{1}{\left(1 - \frac{z}{w}\right)^2} - 1 \right) \\
&= \frac{1}{z^2} + \sum_{w \in \Lambda \setminus \{0\}} \sum_{n \geq 1} (n+1) \left(\frac{z^n}{w^{2+n}} \right) \\
&= \frac{1}{z^2} + \sum_{n \geq 1} (n+1) z^n \sum_{w \in \Lambda \setminus \{0\}} w^{-(2+n)} \\
&= \frac{1}{z^2} + \sum_{n \geq 1} (n+1) z^n G_{n+2} \\
&= \frac{1}{z^2} + \sum_{n \geq 1} (2n+1) z^{2n} G_{2n+2}
\end{aligned}$$

- Consider $f(z) = \wp'(z)^2 - 4\wp(z)^3 + 60G_4\ \wp(z) + 140G_6$, $f(z)$ is necessarily elliptic since it is a combination of elliptic functions. By examining the first terms of the Laurent expansion, we notice that $f(z)$ vanishes at 0. Hence, f has no poles and we conclude that f is constant, and since its constant term is 0 then $f = 0$ so we get $\wp'(z)^2 = 4\wp(z)^3 - 60G_4\wp(z) - 140G_6$. □

Corollary 1.4.27: *Let Λ be a lattice $\langle w_1, w_2 \rangle$, set $w_3 = w_1 + w_2$, we have*

$$\operatorname{div} \wp = -2(0) + (w) + (-w) \quad \text{for some } w \in D$$
$$\operatorname{div} \wp' = -3(0) + (w_1/2) + (w_2/2) + (w_3/3)$$

Notice that the couple $(\wp(z), \wp'(z))$ satisfies the equation $f(x, y) = y^2 - 4x^3 - g_2 x - g_3$ for any z with $g_2 = -60G_4, g_3 = -140G_6$. By a simple change of variable, we can get the equation rewritten in the form $y^2 = x^3 - 4g_2 x - 16g_3$.

Definition 1.4.28 (Curve Of Λ): *Let Λ be a lattice, we define the curve $E = E(\Lambda) \subset \mathbb{A}^2$ given by*

$$E(\Lambda) : y^2 = x^3 - 4g_2 x - 16g_3 \subset \mathbb{A}^2(x, y)$$

with the constants $g_2 = -60G_4(\Lambda)$, $g_3 = -140G_6(\Lambda)$. We define its projective closure with the same notation as the set $E(\Lambda) \subset \mathbb{P}^2$ with

$$E(\Lambda) : y^2 z = x^3 - 4g_2 x z^2 - 16g_3 z^3$$

Lemma 1.4.29: *The curve $E(\Lambda)$ is smooth.*

Theorem 1.4.30: *Let $\Lambda = \langle w_1, w_2 \rangle$ be a lattice of $\mathbb{C}$, the following map*

$$\begin{aligned} \varphi : \mathbb{C}/\Lambda &\to E(\Lambda) \\ z &\mapsto (\wp(z) : \wp'(z)/2 : 1) \\ 0 &\mapsto (0 : 1 : 0) = O_E. \end{aligned}$$

is an isomorphism of Riemann surfaces.

Chapter 2

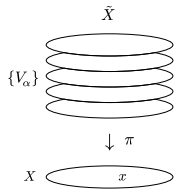
Isogenies Of Complex Tori

2.1. Isogenies

Definition 2.1.1 (Isogeny): *Let T_1, T_2 be two tori, a map $\varphi : T_1 \to T_2$ is said to be an isogeny if it is a morphism of Riemann surfaces and of groups. Denote the set of isogenies from T_1 to T_2 as $\mathrm{Hom}(T_1, T_2)$.*

Definition (Covering Map/Covering Space): *Let $X, \tilde{X}$ be topological spaces. A covering map is a continuous map $\pi : \tilde{X} \to X$ such that $\forall x \in X, \exists U \in \mathcal{N}(x), \exists \{V_\alpha\}_{\alpha \in A}$ pairwise disjoint, $\pi^{-1}(U) = \bigsqcup_{\alpha \in A} V_\alpha$ and $p|_{V_\alpha} : V_\alpha \to U$ is a homeomorphism.*

This space formalizes the specific notion of covering a space X with sheets which are parts of $\tilde{X}$. That is, in a sense, π acts as a projection of $\tilde{X}$ on X sending multiple points of $\tilde{X}$ on X as sheets.



Theorem (Path Lifting): *Let $\pi : \tilde{X} \to X$ be a covering map.*

Lemma 2.1.2: *Let T_1 and T_2 be two tori, $\pi : \mathbb{C} \to T_i$ the canonical projections, $\psi : T_1 \to T_2$ an isogeny, then there exists a unique map $\varphi : \mathbb{C} \to \mathbb{C}$ up to an additive constant $c \in \Lambda_2$ such that the following diagram commutes*

$$\begin{tikzcd} \mathbb{C} \arrow[r, "\varphi"] \arrow[d, "\pi_1"'] & \mathbb{C} \arrow[d, "\pi_2"] \\ T_1 \arrow[r, "\psi"] & T_2 \end{tikzcd}$$

which is equivalent to say that $\pi_2 \circ \varphi = \psi \circ \pi_1$.

Proof. As ψ is an isogeny, then it is a group homomorphism so necessarily $\psi(0) = 0$, define $\varphi(0) = 0$. Notice that $\pi_2 \circ \varphi(0) = 0 = \psi \circ \pi_1(0)$. Let $z \in \mathbb{C}$ and $\tilde{z} \in T_1$ such that $\pi_1(z) = \tilde{z}$, given that $\mathbb{C}$ is path-connected, we can take a continuous path $\gamma : [0,1] \to \mathbb{C}$ such that $\gamma(0) = 0$ nad $\gamma(1) = z$. Define the path $\tilde{\gamma} : [0,1] \to T_2, t = \psi \circ \pi_1 \circ \gamma(t)$, since $\pi_1 : \mathbb{C} \to T_1$ is a covering map then by the path lifting theorem we can lift $\tilde{\gamma}$ to a path $\overline{\gamma} : [0,1] \to \mathbb{C}$ such that $\pi_2 \circ \overline{\gamma} = \tilde{\gamma}$, we set $\varphi(z) = \overline{\gamma}(1)$. $\square$